

EXAM I
PHYS 5103 – Classical Mechanics
75 minutes
September 29, 2026

STUDENT _____

This exam consists of 2 derivations and 1 problem

Derivation 1

Show that for a single particle with constant mass the equation of motion implies the following differential equation for the kinetic energy:

$$\frac{dT}{dt} = \vec{F} \cdot \vec{v}$$

while if the mass varies with time the corresponding equation is:

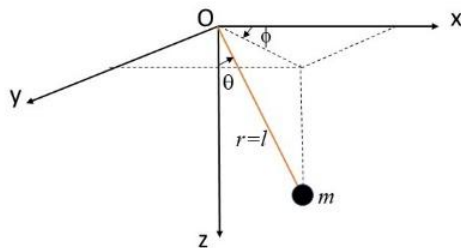
$$\frac{d(mT)}{dt} = \vec{F} \cdot \vec{p}$$

Derivation 2

- (i) Starting with the radial motion equation $\dot{r} = \sqrt{\frac{2}{m} \left(E - V(r) - \frac{l^2}{2mr^2} \right)}$ introduce the quasi-potential $V(r)$ and use the conservation of the total energy E to discuss (a) unbounded, (b) bounded, and (c) circular trajectories for attractive $1/r^2$ forces (i.e. $V(r) = -\frac{k}{r}$, $k > 0$ potentials).
- (ii) For the circular trajectory determine the radius r_0 and total energy E .

PROBLEM 1

Consider a spherical pendulum, a mass m attached to a rod l , as a constrained system with $|\vec{r}| = l$, as shown in the figure.



Write the Lagrangian for the system in spherical coordinates.